

Evolution: build the evidence case

Teacher guide and worked answers

Interpret fossil, anatomical and DNA evidence to justify a claim about evolution and state a limitation.

40 minutes. Use Lesson.html for interactive teaching or Lesson.pptx for editable slides. Slides.pdf is the static alternative. Select one pupil response route and print the shared evidence it requires. Detailed answers below are for staff.

Scheme of work

_passsl/inputs/LAUNCH KS4 - 2026-27.xlsx | LAUNCH Weekly - Spring | C29

Interpret evidence for evolution (data graphs).

Directly targets the SoW evidence and graph outcome. Covers pentadactyl-limb reasoning and data interpretation relevant to Topic 4. This exemplar does not replace separate teaching of named human fossils, stone tools, classification or natural-selection mechanisms.

Prior learning

DNA carries inherited information; species show inherited variation.

Natural selection can change inherited characteristics in populations over generations.

Read a horizontal bar chart and compare numbers.

Success evidence

Identify what an evidence source actually shows.

Use two different evidence types to explain shared ancestry or change over time.

Read the DNA graph accurately and avoid claiming that one living species changed into another.

Equipment

Projector or offline HTML/PPTX; pupil evidence sheet and selected response page.

Pens or mini-whiteboards; optional ruler.

Preparation

Print evidence page for each pupil or pair; select one response route after retrieval.

Keep the answers separate. Pre-teach Ma = million years ago and that a larger Ma value is older.

Explain that Q/A/B/C DNA values are invented practice data; real fossil ages and features are sourced summaries.

Practical care

Paper and screen lesson; no live organisms, biological cultures or chemical work.

Ready to teach alternative

All evidence is supplied for offline use. Discuss, point, annotate or dictate instead of cutting cards or writing extended prose.

Teaching sequence

Slide	Stage	Minutes	Focus
1	Opening	0-2	A whale has finger bones. Why?
2	Retrieval	2-5	Retrieve the tools
3	I do	5-9	Read fossils as evidence
4	I do	9-12	The same underlying plan
5	I do	12-15	Read the DNA graph carefully
6	I do	Within stage	Evolution branches
7	We do	15-22	Museum label rescue
8	We do	Within stage	A second label arrives
9	Hinge	22-25	Which claim is best supported?
10	You do	25-36	Create the museum evidence panel
11	You do	Within stage	A useful explanation structure
12	You do	Within stage	Check your scientific decisions
13	Review	36-39	Compare, improve, hear the response
14	Review	Within stage	What counts as a strong panel?
15	Exit	39-40	One precise final claim

Times are a suggested 40-minute route. Slides marked Within stage share the preceding stage allocation. The optional HTML timer starts only when selected.

1 Opening A whale has finger bones. Why?

Invite one prediction without judging confidence. Say: 'Our job is to connect what we can observe with what it can reasonably tell us.' Share the objective; offer solo or partner route. Do not collect family or personal ancestry examples.

2 Retrieval Retrieve the tools

Allow 30 seconds quiet thought, then simultaneous responses. R1 DNA; genes are sections of DNA. R2 populations over generations. R3 50 Ma, because it means 50 million years before today. If two or more errors, read the vocabulary and model R3 on a number line before continuing.

3 I do Read fossils as evidence

Teacher think-aloud: 'I first read the ages. Pakicetus is older than Dorudon. I next compare the features: walking limbs versus flippers and tiny hind limbs. Together, dated fossils support change in whale relatives through time. I cannot say every fossil species was a direct ancestor.' Ages are approximate ranges summarised from the NHM. The visual simplifies morphology and is not a skeletal reconstruction.

4 I do The same underlying plan

Point in order to one upper bone, two lower bones, wrist and five digits on both limbs. Say: 'I can explain the correspondence as inheritance from a shared ancestor, followed by modification in different lineages.' Five digits is the ancestral pentadactyl pattern; not every living whale species retains the same visible digit count. The chosen schematic represents a five-digit flipper and does not show every small bone or realistic proportions.

5 I do Read the DNA graph carefully

Model: 'The x-axis measures differences, so a shorter bar means greater similarity. Q/A has 2 differences; Q/C has 24. For corresponding DNA this supports Q/A sharing a more recent ancestor. It does not identify the ancestor or date a split.' State clearly: these are constructed sample data, not measurements from real species.

6 I do Evolution branches

Shares the preceding 3-minute I do stage. Briefly trace the two Q/A branches backwards to their nearest shared point. Say: 'This is one possible model; these comparisons support Q/A closeness but do not establish the B/C branching order. Scientists compare all pairs, much more DNA and other evidence.' Do not infer time from branch length.

7 We do Museum label rescue

Run a reasoning clinic: 1 minute private decision; 2 minutes pair or teacher comparison; 2 minutes edit; 2 minutes test the edit against evidence. Accept spoken or written verdicts. Correct verdict: edit/reject the direct-ancestry claim, retain that homologous limb organisation supports a shared ancestor. Dated fossil relatives provide independent evidence of changing limbs. Ask: 'Which precise words go beyond the evidence?'

8 We do A second label arrives

Shares the preceding 7-minute We do stage. Q/A have two sampled differences and do not have identical sampled DNA. Fewer differences support a closer relationship; not descent of one living species from another. Independent sources can strengthen a claim and reveal weaknesses in any one source. Model a concise repaired label after pupils offer alternatives.

Reveal: Q and A are most similar in this sample. This supports a recent shared ancestor; it does not prove A became Q.

9 Hinge Which claim is best supported?

All pupils answer privately before discussion. If B is not secure, circle the sampled quantity and trace the shared branch point again. Challenge 'prove' language. Then release to the suitable independent route.

A. Q evolved directly from living A. - A living tip is not identified as the ancestor of Q.

B. Q and A probably share a more recent ancestor than Q and C. - Fewer differences in corresponding DNA support this relationship, with the sample limitation.

C. Q and C share no ancestor. - More differences do not mean no shared ancestry.

D. A and Q have identical genomes. - They differ at two sampled positions, and whole genomes were not compared.

10 You do Create the museum evidence panel

Set a genuine product: a visitor should leave understanding why evidence supports evolution, without a false direct-ancestor claim. Pupils use the shared evidence page plus one route. At minute 4, check one observation and one inference; at minute 8, check the number and limitation. Do not require all three routes.

11 You do A useful explanation structure

Shares the 11-minute independent stage. Supported route may dictate each sentence while the adult records exactly what was said. Standard pupils choose fossil + anatomical evidence and interpret the graph. Stretch pupils critique the two claims and propose the next investigation.

12 You do Check your scientific decisions

Shares the 11-minute independent stage. Confer with pupils who have given generic 'they look alike' responses: ask them to locate one upper/two lower bones and describe correspondence. For advanced pupils ask whether the short DNA segment samples the whole genome.

13 Review Compare, improve, hear the response

Use a light Lundy loop: provide space for all modes, voice by choice, audience through teacher or partner reading, influence through one recorded revision. Example feedback: 'You identified the same bone plan. Add shared ancestor rather than one living animal becoming another.' Show one strong response from the answer key without naming a pupil.

14 Review What counts as a strong panel?

Shares the preceding 3-minute review. Teacher-authored formative criteria: award one check for each of the three features, not an examination grade. A pupil can demonstrate all three orally or through annotations.

15 Exit One precise final claim

A; 2 differences across 100 sampled DNA positions. Shared ancestry. It cannot establish a direct ancestor sequence, exact split time or full genome similarity from this sample. Collect the exit before the next lesson and reteach only the missing inference.

Answers and assessment

R1-R3 retrieval

R1 DNA; genes are sections of DNA. R2 populations across generations, not an individual changing during its lifetime. R3 50 Ma is older than 33 Ma; Ma means million years ago.

W1 museum label

Edit/reject the direct-ancestry claim. Observation: corresponding limb bones are present. Supported inference: shared ancestry, followed by modification of limb form and function. Humans are not presented as the ancestors of modern whales. Fossil relatives with different combinations of limb features support change through time.

W2 DNA label

Q and A differ at 2 of the 100 sampled positions, so even this sample is not identical. Their greater similarity supports a recent shared ancestor compared with Q/C, not that living A became Q. Whole genomes were not measured.

W3 combine evidence

Independent sources can agree on an explanation and compensate for limitations of a single source; fossil ages/features and corresponding anatomy offer different kinds of evidence. More evidence can also expose conflicts and prompt revision.

Hinge A-D

B is best: fewer differences in corresponding sampled DNA support closer relatedness. A wrongly identifies a living species as the direct ancestor. C wrongly treats larger difference as no ancestry. D ignores two differences and extends a small sample to entire genomes.

S1 / M1 fossil comparison

Pakicetus is older (about 50 Ma) and had walking legs; Dorudon is younger (about 40-33 Ma) with flippers and very small hind legs. Different limb combinations in dated relatives support evolutionary change through time. They do not on their own establish a direct ancestor chain.

S2 / M2 anatomical reasoning

Corresponding one-upper/two-lower bone regions, wrist and digits suggest inherited organisation from a common ancestor, with forms modified in descendant lineages. They are homologous even though function differs. Accept a labelled diagram plus an accurate explanation.

S3 / M3 graph

A is most similar to Q in the sample: 2 differences per 100 aligned positions, compared with 11 for B and 24 for C. Therefore Q/A is the closest supported pair involving Q. Do not generalise to identical entire genomes.

S4 / M4 completed panel and next evidence

Model: "Older whale relatives had walking limbs, while later relatives such as Dorudon had flippers and tiny hind limbs. The shared arrangement of mammalian forelimb bones supports shared ancestry with modification. Fossils are incomplete and these examples do not prove the exact direct ancestor sequence." For M4 accept comparing more corresponding DNA regions, analysing more dated fossil skeletons, or adding independent anatomical evidence with a stated purpose.

T1 corrected claims

Claim 1 overstates completeness and direct descent: the fossil evidence supports change among whale relatives, but does not establish every ancestor or all branches. Claim 2 extends beyond a short sample and ignores two differences: Q/A are similar in those sampled positions; whole-genome identity is not shown.

T2 second DNA region

Yes. Q/A again has the fewest differences (4 rather than 14 or 25), so a second region supports the same closest relationship. The two regions are still samples; evolutionary rates can differ and wider comparative evidence is needed. Do not calculate an exact divergence date.

T3 next check

Accept a testable proposal, for example compare a third corresponding DNA region in Q/A/B/C; predict Q/A will be most similar; agreement raises confidence, disagreement requires examining sampling, alignment or the model. For fossils: seek more securely dated limb material and explain the expected pattern, allowing branches rather than a forced ladder.

T4 integrated caption

Example: "Dated whale relatives show changes in limb form, and corresponding mammalian forelimb bones support shared ancestry. Similarity in comparable DNA provides another line of evidence, although fossil gaps and DNA sampling limit exact reconstructions." Must integrate at least two types with no claim of a living direct ancestor.

E1-E3 exit

E1 Q/A: 2 differences in 100 sampled aligned positions. E2 shared ancestry, followed by modification in descendant groups. E3 accept exact ancestors, an exact split date, complete fossil sequence, or whole genome similarity from 100 positions.

L1 and next step

Answers vary; expect a specific question and an observable revision, such as replacing “proves” with “supports” or adding the sampled quantity. Act on the selected next step; do not grade confidence as knowledge.

Teaching and assessment notes

40 minutes total: opening 2, retrieval 3, I do 10, We do 7, hinge 3, independent 11, review 3, exit 1. Zero-minute slides share their named stage allocation.

This is a focused Spring 1 evidence exemplar, not a complete Topic 4 course. Teach named human fossils, tools, classification and natural-selection examples elsewhere as required by the school sequence.

Keep fact, model and inference separate. The whale ages/features are sourced summaries. The limb drawing is an explanatory schematic. DNA numbers for Q/A/B/C are constructed sample data, designed for comparison rather than a real species claim.

Assessment: securely achieved when the pupil uses two specific observations, gives an accurate explanatory inference, reads one number with its sampled quantity and states a relevant limit. Oral and annotated evidence can meet the same criteria.

Reteach if necessary: cover all but two bars and read the x-axis aloud; then trace living tips backwards to a shared node. Avoid rewarding a long response that claims one modern species turned into another.

TA script: “Point to your observation. Now say what it supports. What would be too strong to claim?” Use 10 seconds processing time before offering the stem.

Access and responsive teaching

Supported

Same scientific objective, with chunked cards, word bank, read-aloud access and short evidence stems. Accept accurately explained oral or annotated responses.

Standard

Use two evidence sources and a reasoned explanation independently.

Stretch

Evaluate limitations, explain a mechanism and transfer to unfamiliar evidence.

Regulation

Offer solo or partner working and a quiet response route. Preview the sequence; allow a brief reset and re-entry at the next numbered task. No compulsory performance or timed competition.

Misconceptions to check

Whales evolved from living humans because both have limb bones.

The shared arrangement supports shared ancestry; neither living group is presented as the ancestor of the other.

Check: Ask pupils to replace the direct arrow between living tips with branches from an ancestor.

One fossil proves the exact whole evolutionary route.

Dated fossils and other evidence are combined; preservation and discovery are incomplete.

Check: Ask what a fossil skeleton cannot show about all earlier populations.

A bar of 24 means 24% of every gene is different.

The graph compares only 100 selected aligned DNA positions in fictional species.

Check: Ask pupils to name the sampled quantity before describing the bar.

Word	Meaning
evolution	Change in inherited characteristics of populations over generations.
fossil	Preserved remains or traces of past organisms.
common ancestor	An ancestral population shared by two or more descendant groups.
homologous	Having a shared evolutionary origin, even when function differs.
inference	An explanation supported by evidence and reasoning.
limitation	A reason the evidence cannot answer every part of a question.

Scientific references

[Pearson Edexcel GCSE Biology 1BI0 specification, Topic 4](#)

Used to check GCSE relevance; teacher-authored practice, not an official assessment. Checked 8 September 2026.

[Natural History Museum: When whales walked on four legs](#)

Approximate ages and broad limb features for three whale relatives; wording and diagrams independently authored. Checked 8 September 2026.

[Smithsonian National Museum of Natural History: Whale Evolution](#)

Checked use of shared anatomy, fossils and branching hypotheses to explain whale ancestry. Checked 8 September 2026.

[NHGRI Comparative Genomics Fact Sheet](#)

Checked why comparing corresponding DNA helps infer evolutionary relationships; no real DNA measurements copied. Checked 8 September 2026.